Engineering Chemistry · Unit_02 · Electrochemistry and corrosion

DRAFT — UE26CY151A · Unit_02 · Electrochemistry and corrosion · Numerical problems based on electrode and cell potential determination

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

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M1

Numerical problems based on electrode and cell potential determination

1 Numerical problems based on electrode and cell potential determination

Study focus: Numerical problems based on electrode and cell potential determination. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000110 · EV0000111

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We'll continue with module four on electrochemical equilibria and in today's class we are going to discuss some numericals on electrochemistry.

Key details

- First we'll look at some problems on Nernst equation and then we look at a problem on ion selective electrode that is the glass electrode.
- And the values in the parenthesis stand for the concentrations of Fe2 plus ions and AG plus ions.
- The question says write the overall cell reaction and determine E naught cell and E cell at 25 degrees centigrade.
- Standard reduction potential of the Fe two plus Fe electrode that is minus 0.44 volts and the standard reduction potential of AG plus AG that is 0.80 volts.
- First and foremost we need to write the overall cell reaction, iron being at the anode.
- We have to write the reaction as Fe giving you Fe2 plus 2 electron and this being silver being at the cathode, the reaction will be reduction reaction.
- We have multiplied this reaction with 2.
- In order to balance the number of electrons, overall reaction is Fe plus 2A G plus is giving you Fe2 plus 2 ag.
- We know the Nernst equation.
- So from the Nernst equation we will be able to determine E cell because we know what is E naught cell.

- But first and foremost, whenever you have a reaction, when the overall reaction, it's always a good practice to find out what is Q.
- Before you go ahead and substitute in the Narnst equation as done here.
- We have Fe2 plus and AG the whole square that is concentration of the reactants by Fe and AG plus square.
- Now we have not taken AG and FE into consideration because we know that for pure substances.
- The concentration is taken as 1.
- And the answer we get is 1.2193 volts.
- This potential, cell potential at 25 degrees for a concentration cell.
- One is the electrode type, the other one is the electrolyte type.
- We see that the electrolyte concentration is just one, but the activity of the electrode itself is different.
- We are calling it electro type electrode type because here the activity of hide the partial pressure of hydrogen is 8 atmosphere.
- And here the partial pressure of hydrogen at the cathode is 2 atmosphere.
- We know from the expression E cell is equal to 0.0591 by n log of partial pressure of hydrogen at the anode by partial pressure of hydrogen at the cathode.

Example or application from the lecture

- So the first question is we have been given a cell where we have coupled the iron electrode with the silver electrode.
- What is given to us is the.
- Let us find out the E naught cell, which is given by E naught of the cathode minus E0 of the anode.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000110 · EV0000111 · EV0000112 · +1 source units in pack.json

Core explanation · 05:00–09:59

At the cathode, this is the reaction.

Key details

- At the anode, the reverse reaction would be taking place.
- H2 is giving you 2H plus 2 electron that is at the anode.
- And the cathode partial pressure would be 2 atmosphere.
- When we find out the cell potential expression, it comes out as 0.0591 by n log of partial pressure of hydrogen at anode by partial pressure of hydrogen at the cathode.
- And understand here E0 is not there because E0 is same for the anode as well as the cathode and hence cancels out.
- And the answer now is 0.1779 volts.
- What do you understand by decinormal calomel electrode?
- We know that the.

- Potential of the calomar electrode depends on the chloride ion concentration or the KCl concentration which is used.
- A saturated calomel electrode.
- That forms the anode.
- We are asked to write the cell representation and calculate the concentration of Cl minus IN the saturated calomel electrode.
- Please understand that we already know that the cathode concentration of KCl is 0.1 molar or normal.
- And we need to determine the chloride ion concentration for the saturated calomel electrode.
- Anode, it is a saturated calamar electrode for which the concentration is not known.
- The decinormal calomel electrode, and for any calomal electrode, we know the expression is E naught.
- Minus 0.0591 log of the concentration of Cl minus.
- Now here looking at the cathode, it's a decinormal calomyl electrode, therefore Cl minus is 0.1, and here it is.
- X because we want we don't know what is the concentration of Cl minus.
- Calomyl electrode when it is a decenormal calomal electrode, the Nernst equation for the decenormal calomal electrode minus the Nernst equation for the saturated calomel electrode.
- Therefore, when we substitute we get 0.0988 is the e cell divided by 0.0591 log of x by point one.
- And therefore, when we substitute the values.
- We get we need to take antilog over here.
- When we do the antilog, we find that x is equal to 4.69 molar.
- Come to the next problem now.
- Here we have silver chloride 0.1 molar concentration of chloride ion.
- Can we identify this is a silver chloride electrode.
- Which is a DC normal or having a concentration of 0.1 normal or 0.1 molar chloride concentration.
- KCl concentration or HCl concentration is 0.1 molar.
- And the other side we have Fe two plus.
- Which has a concentration 0.29 molar, Fe 3 plus, which has a concentration 0.18 molar, and this is in contact with an inert electrode platinum.
- If you recognize what is this type of electrode, where we have.

Example or application from the lecture

- Let us look at a third type of problem where we have a decinormal calomyl electrode.
- First, let us write the cell representation.

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Core explanation · 10:00–14:59

Both the oxidized and the reduced species present together, this is a redox electrode.

Key details

- It says write the half cell reactions and overall cell reaction.
- And calculate E naught cell and E cell at 298 Kelvin.
- What is the solution?
- So first we need to write the half cell reactions.
- What would be the oxidation reaction for silver chloride electrode?
- We have already learned it.
- We will have the reaction Fe3 plus taking up electron and becoming Fe2 plus.
- That is the reduction reaction.
- Whenever we have such a question.
- It's always a good practice to write the half cell reactions and then the overall cell reaction, even though it is the question just says write the overall cell reaction.
- And next, the first thing we should do now is to find out what is Q because this has to be done very carefully.
- F E two plus by Cl minus.
- Fe three plus.
- Cl minus and Fe 3 plus.
- That is the value of Q.
- Yes, that would be E naught cathode minus E0 anode.
- We got E naught cell.
- And N I know is 1 because there's only one electron getting exchanged in the overall reaction also.
- And now I have an expression here that is the Q expression.
- How did I get this?
- This I got from the overall cell reaction.
- We are the product concentration divided by the.
- Reactant concentration and therefore we find if we substitute 0.548 is the E naught cell and for Q we substitute the values 0.29, 0.18 and 0.1.
- And we find that.
- The overall cell E cell is 0.4767 volts.
- So that means we are straight away looking at a concentration cell.
- This is now we can make out the electrolyte concentrations are different.
- Therefore, this is an electrolyte type of concentration cell.
- Three plus three electrons is giving you.
- AU and this will happen at the cathode and at the anode we have the reverse that is AU is giving you AU3 plus three electrons.
- It is three.
- Therefore, if we substitute three here.
- And we know the concentrations at the cathode and the anode, we find that since n is equal to 3, E cell comes out to be 0.00749 volts.
- What you have to understand here is that whenever we do concentration cells as explained.

Example or application from the lecture

- Given the standard reduction potential values of both the electrodes are given to us, R is 8.314 joules per Kelvin per mole per mole and f is equal to nine six five zero faraday is nine

six five zero coulombs per mole.
- Let us see.
- The values are given to us.

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Core explanation · 15:00–19:59

During the when I was talking about concentration cells teaching you, e cell will be a very small value.

Key details

- You can see here E cell is a very small value, that's because the main contribution to e cell comes from E naught cell and E0 cell.
- From the cathode and anode being same as E naught of the anode and cathode being same, so E0 cell comes out to be zero.
- E0 for anode and cathode, they've got cancelled, therefore there is no E0 cell at all, and hence the concentration is.
- The cell potential is 0.00749 volts.
- That means it's a very small value.
- And since both the oxidized and the reduced species are present together in the same solution and an inert wire is placed to collect the potential, this is a redox electrode.
- We have to write the cell representation, write the reactions at the electrodes, and we have to calculate E naught cell and E cell at 298 Kelvin.
- Here we have to write the cell representation.
- Therefore this we place at the anode and this will be the cathode.
- The reactions at anode would be C U plus is giving you CU2 plus 2 electron and here at the.
- Cathode we have this reaction, the calom electrode is getting reduced.
- Hg2Cl2 plus 2 electron is giving you 2 H G plus 2 Cl minus.
- Overall reaction would be 2CU plus H2Cl2 is giving you 2 C U2 plus 2 Hg plus 2 Cl minus.
- E0 cell will be E cathode, E0 of the cathode minus E0 of the anode, that is 0.281 minus 0.153, and the value for E0 cell is 0.128 volts.
- E no E cell.
- For E cell I need Q.
- So first itself, so C U2 plus using the overall reaction.
- C U2 plus squared.
- And here it is Cl minus.
- squared and then that is divided by.
- C u plus square.
- And so we substitute that expression over here.
- And n in this case we can see is two electron.
- of C U2 plus Cl minus and C U plus and these are substituted here and N is equal to 2.
- We have to write the E naught cell and E cell at 298 Kelvin.
- E naught for the gold electrode is 1.52 volts and E naught.

- For an iron electrode is minus 0.44 volts.
- F is nine six five zero coulombs per mole.
- We know the half cell reactions that is Fe iron is undergoing oxidation, so Fe 2 plus 2 electron, and here we have AU3 plus 3 electron at the cathode undergoing reduction A giving you AU.

Example or application from the lecture

- Let us look at the next problem where we have a decommal calomel electrode and it is used to determine the potential of the following redox electrode.
- So given the standard reduction potential for the calomar electrode that means that the chloride concentration is 1 molar and therefore this is 0.281 volts and C U2 plus C U, the standard reduction potential is given as 0.153.
- Let us look at the solution.

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Core explanation · 20:00–24:59

That's exactly what we have done here, and the overall reaction now we get 3 FE plus 2AU3 plus, giving you 3 Fe plus.

Key details

- 2 plus 2 AU.
- That is the net cell reaction.
- As I mentioned before, always good to write the Q first before proceeding with the calculation.
- If it was not at 298 Kelvin, then we would require these values.
- Because in that case, what is the expression you would use?
- 2.303 R T over NF.
- By AU plus 3 squared.
- That's exactly the Q value here.
- What is N now here?
- has this will become 6, this will also become 6, therefore the number of electrons are going to cancel each other.
- And then we get n as 6.
- And when you substitute the substitute the values, this is the expression we get and the final answer we get it as.
- One point nine eight three six votes.
- which is not known to us.
- The cell potentials are 0.215 volts and 0.385 volts, which are when the electrode is in contact with the solution of pH is equal to 7 and with the solution of unknown pH.
- E cell plus 0.0591 PH plus E saturated calomol electrode.
- So this expression first will help us to determine E naught G because E naught G value is not known to us unless we use.

- A solution of known pH.
- What is the pH here?
- So first and foremost we find out what is E0 G, and that is 0.8727 volts, and then we use this expression.
- P H is equal to E naught G minus E SE minus E cell divided by 0.0591.
- And E saturated calomin electrode is known to us, and E cell measured.
- Now we have to use this value because now the pH is unknown.
- Here, what is the E cell we use?
- Because this is the pH seven, we know the pH seven.
- T cell we are going to use now.
- When we do that, we will substitute these values and what we get is ph is equal to 4.12.
- Module of electrochemical equilibria.
- Now in the next class we will start with unit three, that is on energy storage devices.

Example or application from the lecture

- And let us write the Q again.
- Now let us find out the E cell as that is what we is to be determined E cell at 298 Kelvin, therefore 0.0591.
- Let us come to the last problem.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

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A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Numerical problems based on electrode and cell potential determination transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__8150689d77526239.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Numerical problems based on electrode and cell potential determination transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__8150689d77526239.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Numerical problems based on electrode and cell potential determination transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__8150689d77526239.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9988; only the JSON segmentation supplies the displayed timestamps.
Numerical problems based on electrode and cell potential determination transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__8150689d77526239.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00022	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...
SRC00023	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00024	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...
SRC00025	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...

C Revision and retrieval practice